CW
27/07

L'Hopital's Rule

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Indeterminate forms : $\frac{0}{0}$ - $\lim_{x\to a}\frac{f(x)}{g(x)} = \frac{0}{0} = \frac{f'(x)}{g'(x)}$

$$\lim_{x\to 1}\frac{\ln x}{x-1} = \lim_{x\to 1}\frac{\frac{1}{x}}{1} = \lim_{x\to 1}\frac{1}{x} = 1$$

$$\lim_{x\to 0}\frac{\tan x - x}{x^3} = \lim_{x\to 0}\frac{\sec^2 x - 1}{3x^2} = \lim_{x\to 0}\frac{2\sec x \sec x\tan x - 0}{6x}$$

$$= \lim_{x\to 0}\frac{2\sec^2 x\tan x}{6x} = \frac{1}{3}\lim_{x\to 0}\frac{\sec^2 x\tan x}{x}$$

$$= \frac{1}{3}$$

$\frac{0}{0}$

$$\lim_{x\to\infty}\frac{x}{e^x} = \lim_{x\to\infty}\frac{1}{e^x} = \lim_{x\to\infty}\frac{0}{e^x} = 0$$

$$\lim_{x\to 0}\frac{\ln x}{\operatorname{cosec} x} = \lim_{x\to 0}\frac{\frac{1}{x}}{-\operatorname{cosec} x\cot x}$$

$\frac{\infty}{\infty}$

$$= \lim_{x\to 0}\frac{\tan x}{\frac{x}{\sin x}} = \lim_{x\to 0}\frac{\sin x}{x}\cdot\lim_{x\to 0}\tan x$$

$$= 0$$

$0.\infty = \frac{\infty}{\frac{1}{0}}$

$$\lim_{x\to 0}\frac{\ln x}{\frac{1}{x}} = \frac{\frac{1}{x}}{-x^{-2}} = \lim_{x\to 0}\frac{x^{-1}}{-x^{-2}} = \lim_{x\to 0} -x$$

$$= 0$$

Case 1 '+' :
Default :

$$\lim_{x\to\infty} x^3 e^{-x^2} = \lim_{x\to\infty} \frac{x^3}{e^{x^2}} = \lim_{x\to\infty} \frac{3x^2}{e^{x^2}\cdot 2x}$$

$$= \lim_{x\to\infty} \frac{6x}{e^{x^2}\cdot 4x^2} = \frac{6}{4}\lim_{x\to\infty} \frac{1}{4x\cdot e^{x^2}} = \frac{6}{4}\lim_{x\to\infty} \frac{-x^{-2}}{e^{x^2}}$$

$$= \frac{6}{4}\lim_{x\to\infty} \frac{-1}{2xe^{x^2}} = 0$$

$$\lim_{x\to 0^+}\left(\frac{1}{x} - \frac{1}{\sin x}\right) = \lim_{x\to 0^+} \frac{\sin x - x}{x\sin x}$$

$\infty - \infty$
$(+\infty) - (+\infty)$
$(-\infty) + (-\infty)$
$(-\infty) + (+\infty)$

$$= \lim_{x\to 0^+} \frac{\cos x - 1}{\sin x - x\cos x} = \lim_{x\to 0^+} \frac{-\sin x}{\cos x - x\sin x\cos x} = \frac{0}{2-0} +$$

$$\lim_{x\to a}\left[e^{\ln f(x)}\right]^{g(x)} = \lim_{x\to a} e^{g(x)\ln f(x)}$$

$0^0, \infty^0, 1^\infty$
$\lim_{x\to a} f(x)^{g(x)}$

$$= e^{\lim_{x\to a} g(x)\ln f(x)}$$

Ex

$$\lim_{x\to 0}(1+\sin x)^{\frac{1}{x}} = \lim_{x\to 0}\left[e^{\ln(1+\sin x)}\right]^{1/x}$$

$$= \lim_{x\to 0} e^{\frac{1}{x}\ln(1+\sin x)} = e^{\lim_{x\to 0}\frac{1}{x}\ln(1+\sin x)}$$

$$= e^{\lim_{x\to 0}\frac{\cos x}{1+\sin x}} = e^{1} = e$$

$$\lim_{x\to 0} x^{\sqrt{x}} = \lim_{x\to a}\left[e^{\ln x}\right]^{\sqrt{x}} = \lim_{x\to a} e^{\sqrt{x}\ln x}$$

$$= e^{\lim_{x\to a}\sqrt{x}\ln x} = e^{\lim_{x\to a}\frac{1}{2\sqrt{x}x}} = e^{\frac{1}{2}\lim_{x\to a}} =$$

$$= e^{\lim_{x\to 0} \frac{1}{2\sqrt{x}\,x}} = e^{\frac{1}{2}\cdot\frac{2\sqrt{x}}{x}} \quad e^{\frac{1}{2}\cdot\frac{2}{2\sqrt{x}}}$$

$$= e^{\lim_{x\to 0} \frac{\frac{1}{x}}{-\frac{1}{2}x^{-\frac{3}{2}}}} = e^{\lim_{x\to 0} 2x^{-\frac{1}{2}}} = e^0 = 1$$

$$\lim_{x\to 0} x^{\frac{1}{x}} = e^{\lim_{x\to 0}\left[e^{\ln x}\right]^{\frac{1}{x}}} = e^{\lim_{x\to 0}\frac{1}{x}\ln x}$$

$$= e^{\lim_{x\to 0}\frac{-x^{-2}}{x}} = e^{\lim_{x\to 0} -x^{-3}} = e^0 = 1$$

$$= e^{\lim_{x\to 0}\frac{\ln x - 1}{x^2}} = e^{\lim_{x\to 0}} \qquad \boxed{e^{\infty} = \infty}$$

29 $\lim_{x\to\infty}(x - \ln x) = \lim_{x\to\infty}$ $1 - \frac{1}{x}$

$$= \lim_{x\to\infty}\frac{x-1}{x} = 1 - \lim_{x\to\infty} -x^{-2}$$

$$\lim_{x\to\infty}(\ln e^x - \ln x) = \lim_{x\to\infty}\ln\left(\frac{e^x}{x}\right)$$

$$= \ln \lim_{x\to\infty}\frac{e^x}{x} = \ln \lim_{x\to\infty} e^x = \infty$$

25 $\lim_{x\to 1}\left(\frac{x}{x-1} - \frac{1}{\ln x}\right) = \lim_{x\to 1}\frac{x\ln x - x + 1}{(x-1)\ln x}$